AI-Driven Delinquency Prediction: Business Insights Report

1. Predictive Insights

High-Risk Customer Segments
• Customers with repeated late or missed payments represent the highest-risk segment because historical payment behavior is strongly associated with future delinquency.
• Customers with high credit utilization and high debt-to-income (DTI) ratios are more likely to become delinquent due to financial stress.
• Customers with unstable employment status, particularly unemployed customers, show higher delinquency rates because of reduced repayment capacity.

Key Variables Influencing Delinquency
EDA and predictive modeling identified the following variables as the strongest predictors:
1. Credit Utilization
2. Debt-to-Income Ratio
3. Payment History (late and missed payments)
4. Income
5. Employment Status

Top 3 Risk Factors
• Repeated late or missed payments.
• High credit utilization levels.
• Employment instability combined with high debt burden.

2. SMART Business Recommendation

Insight: High credit utilization significantly increases delinquency risk.

SMART Goal:
Reduce delinquency among customers with credit utilization above 80% by 15% within the next 12 months by implementing proactive outreach, personalized repayment plans, and early-warning alerts.

Stakeholder Rationale:
Targeted interventions for customers experiencing financial stress can improve repayment behavior, reduce default losses, and strengthen customer relationships while supporting Geldium's objective of proactive risk management.

3. Ethical and Responsible AI Considerations

Fairness Risks and Mitigation
Risk 1: Certain demographic groups may receive disproportionately higher risk scores due to historical bias in training data.
Mitigation: Regularly audit model performance across customer groups and rebalance training data where necessary.

Risk 2: False positives may incorrectly classify low-risk customers as delinquent, resulting in unfair treatment.
Mitigation: Continuously monitor false positive rates, review model thresholds, and include human oversight for high-impact decisions.

Explaining Predictions to Non-Technical Stakeholders
Model predictions can be explained in plain language by highlighting the key factors influencing each risk score, such as payment history, credit utilization, and debt levels, rather than focusing on technical model details.

Responsible AI Practices
The approach promotes transparency by using an interpretable Logistic Regression model, conducting fairness checks, documenting assumptions, and performing periodic model reviews to ensure ongoing

reliability and accountability.

Conclusion

The proposed AI-driven delinquency prediction framework enables Geldium to identify high-risk customers early, prioritize interventions, and reduce delinquency while maintaining responsible and transparent AI practices.